

Senior Software Engineer – Geospatial & Mobility Platform

Location: Hanoi / Ho Chi Minh City (*Requires initial 6–9 months on-site in Hanoi*)

Focus: Mapping Core, Dynamic Pricing Engines, and Marketplace Matching Systems

1. THE ROLE

As a foundational hire, you will own the technical architecture that bridges geospatial routing physics with real-time marketplace economics. You will build high-performance mapping pipelines alongside the stateful and stateless microservices that power driver discovery, ETA calculations, live traffic monitoring, dynamic surge pricing, and batch-matching dispatch engines.

2. KEY RESPONSIBILITIES

- **Stateless Mapping Engines:** Design high-compute, low-latency services for O(R) complex queries, including Location Search, Geocoding, and Route/Time (ETA) Estimation.
- **Stateful Marketplace Economics:** Build and scale real-time, stateful services tracking driver supply and rider demand (O(D) state size) to power dynamic surge pricing and toll/tax calculations.
- **Feature Pipeline & Matching:** Architect data flows that flatten route, rider, and driver metrics into a consolidated feature store, feeding a high-compute batch Matching Engine.
- **Scale with Go (Golang):** Transition CPU-heavy, ultra-low-latency backend components from Node.js/Python into Go to manage real-time GPS streaming and high-concurrency read requests.

3. REQUIREMENTS

Must-have

- **Experience:** Minimum 5 years of backend engineering experience with a proven track record in 0-to-1 system design and microservices architecture.
- **Routing Engine Expertise:** Deep, hands-on experience integrating, tuning, and deploying open-source routing engines (OSRM, Valhalla, or GraphHopper) to handle real-time distance and time estimation queries.
- **Mobility Marketplace Background:** Direct experience managing the tradeoffs between Stateful vs. Stateless services in an on-demand ecosystem (e.g., handling real-time data staleness for driver location, surge pricing, or traffic updates).
- **Tech Stack:** Advanced proficiency in Node.js and/or Python; strong aptitude and

readiness to immediately write and deploy production-grade Go (Golang).

- **Infrastructure & Streaming:** Experience with caching and streaming layers (Redis, Kafka) to process high-throughput real-time pings and handle concurrent read requests.
- **Mobility:** Ready to relocate to Hanoi for the first 6–9 months to build the foundational architecture on-site with the core team.

Nice-to-have

- Familiarity with PostGIS, spatial indexing (H3/S2 geometry), and planet-scale OSM tooling (osm2pgsql, Osmium).
- Experience with search/autocomplete optimization using Elasticsearch/OpenSearch.
- Prior background at a major ride-hailing, food-delivery, or fleet-logistics platform.

4. WHAT WE OFFER

- Full relocation support for your initial phase in Hanoi and a dedicated learning/training budget for rapid Go (Golang) mastery.
- Competitive senior salary, performance bonuses, 13th-month pay, and premium health insurance.
- The rare opportunity to write the foundational code for a national-scale digital infrastructure project.